

Title: Fault Injection Detection in Deep Neural Networks Using Multi-Layer Internal States and Support Vector Data Description

Problem Statement

Deep Neural Networks (DNNs) are increasingly deployed in applications where reliable predictions are critical. One problem is that attackers can inject faults into the model resulting in adversarial bit flips in model weights or biases, can affect internal calculations and cause the model to make incorrect predictions. Finding the effects of these faults is challenging since their impact depends on the specific parameter, bit position, and network layer affected. Some faults substantially change the final prediction, whereas others induce smaller internal deviations that remain unnoticed. Many existing approaches rely mainly on the final output. Yet a fault can influence internal states across multiple layers as it propagates through the DNN. Monitoring several layers may therefore better capture abnormal behavior and enable earlier fault detection, before effects are clearly visible at the output. This thesis investigates whether multi-layer internal states can distinguish normal DNN execution from fault-injected execution.

Objective:

The main objective of this thesis is to determine whether multi-layer internal states can reliably detect fault-injection attacks in DNNs. The study will model normal layer-wise behavior and use SVDD-based anomaly detection to flag deviations, evaluating detection performance and overhead across architectures and fault models. A secondary objective is to quantify whether fused multi-layer monitoring outperforms per-layer monitoring and to identify minimal monitoring configurations that preserve high detection accuracy.

Proposed Technical Directions:

The thesis will extract compact, informative features from selected internal layers of a DNN during both normal and fault-injected inference. Faults will be introduced primarily as bit flips in weights and biases, varied across layers and bit positions to cover both obvious and subtle error patterns.

Support Vector Data Description (SVDD) will be trained on clean executions to model expected layer-wise behavior and to flag deviations indicative of faults. Two complementary directions will be pursued: (1) fuse multi-layer features into a joint representation for anomaly detection, and (2) train per-layer SVDD models to localize where anomalies first emerge and how they propagate.

Evaluation will span multiple fault-injection scenarios and models. Performance will be quantified with detection rate, false-positive rate, precision, recall, F1-score, and ROC-AUC, alongside runtime and memory overhead. Results will determine whether multi-layer internal-state monitoring yields a measurable advantage over single-layer monitoring for detecting fault-injection attacks.